6 Spatiotemporal Cortical Dynamics—The Wilson Cowan Equations

It is hoped that the relative simplicity of the model may serve as a basis for a better understanding of the functional significance of cortical complexity.
—Hugh Wilson and Jack Cowan, 1973 [WC73]

6.1 Before Wilson and Cowan

So far, we have focused on models that represent the excitability of single cells (or components of single cells). Brain dynamics emerge from the interactions of single cells—in the case of vertebrates, the interactions between millions and billions of those cells. So at first glance, modeling the complexity of the patterns of activity from this complex network seems daunting. Nevertheless, it was not long after Hodgkin and Huxley's work with the squid axon that such larger-scale brain dynamics were seriously addressed:

> Cells having some properties similar to those of neurons are considered. A mass of such cells, randomly placed together with a uniform volume density, appears capable of supporting various simple forms of activity, including plane waves, spherical and circular waves and vortex effects. The propagation of a plane wave of activity has been considered in some detail. (R. L. Beurle, 1956 [Beu56])

If it were not for the fact that the date on the above quotation were 1956, these sentences would fit well within any number of papers published recently on the theory and experimental findings of cortical activity.

Beurle was interested in describing the properties of a mass of cells in a brain. He understood clearly that through their interactions, the properties of the mass would be distinct from the activities of cells in isolation. Perhaps they would bear similarities to the activities that drive organismal behavior.

There is an implicit recognition of *emergence* throughout Beurle's writing. This rather attractive word, with a rather faddish emergence in scientific circles over the past decade, probably has no better description than Ablowitz's 1939 paper [Abl39]. Incredibly, one can find this concept, as *emergents*, used in the modern sense in Lewes's 1875 book, as properties that cannot be deduced by the sum of its parts:

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No one supposes that the properties of each element combined in water could be deduced from the observed properties of the combination; no one supposes that [from] the observed properties of oxygen and hydrogen, separately considered, the properties of water could be deduced. (G. H. Lewes, 1875 [Lew75])[1]

An ensemble of linear dynamical systems, coupled linearly, will produce macroscopic dynamics readily accounted for by the (linear) summation of the individual dynamics. Nonlinear systems, on the other hand, can have emergent properties whereby the whole behaves differently than the sum of its parts would have suggested. Neurons, with thresholds, are paradigmatic examples of nonlinear networks where we expect emergent dynamics to emerge.

Beurle assumed that the behavior of the mass of cells (idealized in his formulation) would have a threshold.

He assumed that these model neurons would have very long refractory periods, so that once they fired, in some kind of pattern of activity, they did not fire again during that pattern. Figure 6.1A shows a traveling wave consuming sensitive cells and turning them refractory in its wake.

In figure 6.1B Beurle shows waves colliding and reflecting from this encounter to go back along the tracks of the incident waves. In figure 6.1C, he predicts a wave encountering a cortical boundary and refracting as its propagation speed at the boundary slows. Fifty years later, Xu and colleagues would find this phenomenon in the compression and reflection of waves of activity in visual cortex as they transit from visual cortical regions V1 to V2 [XHTW07]. Such waves compress, as would light crossing an analogous boundary from an area of low to one of higher refractive index. And along with the compression, reflected brain waves are observed [XHTW07]. Many of these wave properties in visual cortex were envisioned in Beurle's early insight.

6.2 Wilson and Cowan before 1973

Hugh Wilson was a postdoc with Jack Cowan in the early 1970s. Jack had an unusual legacy for a neuroscientist. His mentors included Claude Shannon and Dennis Gabor—giants in the mathematics and theory of communication (Shannon) and signal processing and holography (Gabor). His thinking about the brain was influenced by his association with Norbert Wiener, Warren McCulloch, and Walter Pitts, a pivotal group of scientists in the mid-twentieth century who initiated efforts at fusing engineering and mathematics with our

1. Lewes was the lover of Marian Evans, who wrote under pen name George Eliot in Victorian England. Upon her death, Evans established the George Henry Lewes Trust at Cambridge to support students studying physiology. Among the list of students so supported who became central figures in twentieth-century science included Nobel laureates H. H. Dale, C. S. Sherrington, and A. V. Hill. Importantly for our purposes here is that W. A. H. Rushton was supported by the Lewes Trust when he performed the seminal work upon which our chapter 11 is based.

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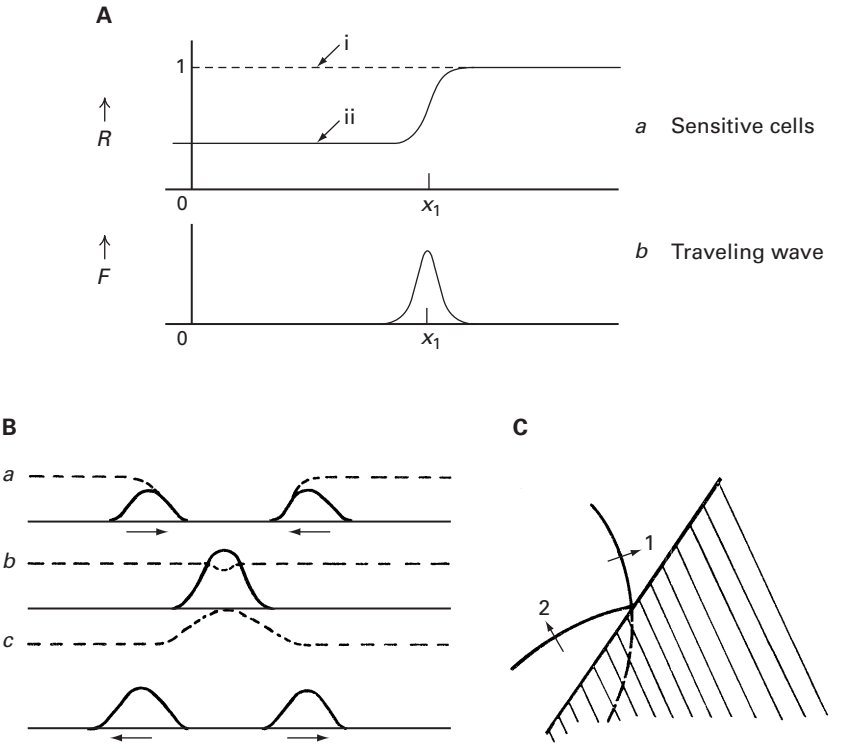


Figure 6.1
Traveling wave (A), reflected waves (B), and interaction with boundary (C). Modified from [Beu56] with permission.

understanding of the brain (exemplified in [MP43, Wie48, LMMP59]. The collaboration of Wilson and Cowan produced two seminal papers in computational neuroscience from their collaboration. In the first [WC72], they sought a deterministic model of cortex.[2]

Their quest was to develop a deterministic model of cortex that used mean values from the statistical processes of the underlying cellular spiking. There are no spikes in their analysis. They discuss the local redundancy in cortex. For the physicists, this is essentially a local mean field approach to looking at cortical dynamics, and if they were doing this a decade or two later, they might have talked about renormalizing the individual cells into larger, more equivalent cells in cortex. *It is just such local redundancy which must be invoked to justify characterizing spatially localized neural populations by a single variable* [WC72].

Wilson and Cowan made a major break from Beurle by assuming that there were excitatory and inhibitory cells that were ubiquitous in the brain. I will take that assumption

2. A detailed personal history and discussion of this work can be found in [Cow10].

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a little further: Composites of interacting excitatory and inhibitory cells are ubiquitous in spinal cord, brainstem, cortex—in every component of the vertebrate central nervous system. *All nervous processes of any complexity are dependent on the interaction of excitatory and inhibitory cells* [WC72]. Wilson and Cowan realized that such excitatory-inhibitory interplay would be essential to establish dynamical stability of the brain. Indeed, J. C. Eccles would suggest that the entire purpose of inhibitory neurons might be to prevent seizures (personal communication, 1985). This principle of the balance of excitation and inhibition has taken on increasing importance following the experimental work on *up* states [SHM03, HDHM06]. And such excitatory-inhibitory interplay will play a crucial role later in our efforts to reconstruct epileptic network dynamics (chapter 12).

Dale's principle says that one neuron secretes only one type of transmitter—that "the chemical function, whether cholinergic or adrenergic, is characteristic for each particular neurone, and unchangeable" [Dal35]. This required Wilson and Cowan to separate independently their populations of cells, because some populations are only excitatory and some are only inhibitory. This is an *independence principle*, which is in part analogous to the independence of the ion channels that Hodgkin and Huxley applied to nerve membrane dynamics (chapter 3). In a qualitative sense, sodium is excitatory and potassium is inhibitory in a patch of the neuron membrane, but one cannot carry that analogy too far. Nevertheless, we are about to discuss nonlinear wave propagation and nonlinear pattern formation in cortex, and those processes do share dynamical properties with the mechanisms of excitability and propagation of single-action potentials.

Wilson and Cowan also assumed that every kind of interconnection that is possible between these two types of cells would be present. This lets their model be general enough to encompass all possible ways that excitatory and inhibitory populations can interact. By setting the connection strengths and parameters of the model, it could represent all possible topologies that cortex might contain.

They assumed that the excitatory-to-excitatory connections were tighter and more localized than the reach of inhibitory connections. The key variable they use is Beurle's—the proportion of cells that act per unit time, and that will be included in the following integral formulations they propose.

Let $E(t)$ be the proportion of excitatory cells active per unit of time. Similarly, $I(t)$ will be the proportion of inhibitory cells active per unit of time. $E(t) = 0$ and $I(t) = 0$ will be the rest state of the subpopulations when their background activity is low. Note that this gives us the prospect of *negative* values of E or I that are meaningful—the activity is depressed relative to baseline.

If the absolute refractory period is r, the proportion of excitatory cells E that are refractory could be obtained by integrating from $t - r$ to the present time t

$$\text{Proportion refractory} = \int_{t-r}^{t} E(t')dt'$$

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If those are the proportion of cells that are refractory, then the ones that are sensitive, the ones Beurle had talked about, are 1 minus this fraction.

$$\text{Proportion sensitive} = 1 - \int_{t-r}^{t} E(t')dt' \tag{6.1}$$

Wilson and Cowan sought a *subpopulation response function*, S_e for excitatory (or S_i for inhibitory), that would represent the proportion of the subpopulation receiving at least threshold activation per unit time. This is the proportion of cells in a subpopulation that *would* respond to a given level of excitation if none of them were refractory at that time.

They next assumed that the population has a distribution D of *thresholds* θ, $D(\theta)$, which might be represented by a Gaussian distribution. It did not really matter what the distribution was, but it had to have a hump, and it had to be focused around a mean distribution of thresholds. If all these cells have the same average excitation $x(t)$, then the fraction of these cells S that were over the threshold at time t would be

$$S(x) = \int_{0}^{x(t)} D(\theta)d\theta \tag{6.2}$$

Such a formulation led to sigmoid activation curves. An alternative way to formulate this would be to assume that the thresholds of all cells were the same, but that they received a different number of synapses distributed according to a similar one-humped distribution function.

By the time of Wilson and Cowan's work, it was well known that a neuronal population might have a definite threshold for excitation, and that the input-output relation often appeared to be of sigmoidal shape (figure 6.2).

Wilfrid Rall's studies on the quantification of the recruitment within the motoneuron pool sought to place such sigmoidal activations on a sound quantitative footing.[3] Rall was interested in addressing how the spatial summation of synaptic excitation interacted with the distribution of thresholds within the motoneuron pool to produce the amplitude of the monosynaptic reflex output [Ral55a]. His statistical theory to account for these findings explained how simple Gaussian activation functions, shown in figure 6.3A, combined with thresholds for neuronal activation within the population, would generate sigmoidal population activation curves, shown in figure 6.3B.[4]

3. Rall would later spend the bulk of his career at the Mathematical Research Branch at NIH for several decades, performing seminal work in the physics of neurons. But his previous doctoral thesis was done in Dunedin with J. C. Eccles, and his subsequent work in Dunedin is reflected in the results shown here.

4. The nature of correlated and uncorrelated fluctuations seen in these input-output curves became of renewed interest when the surgical treatment of children with spasticity involved measurement of reflex input-output strengths in selective dorsal rhizotomy [WS93]. A more detailed analysis of fluctuations in single cell and population activity can be found in [GFK+94].

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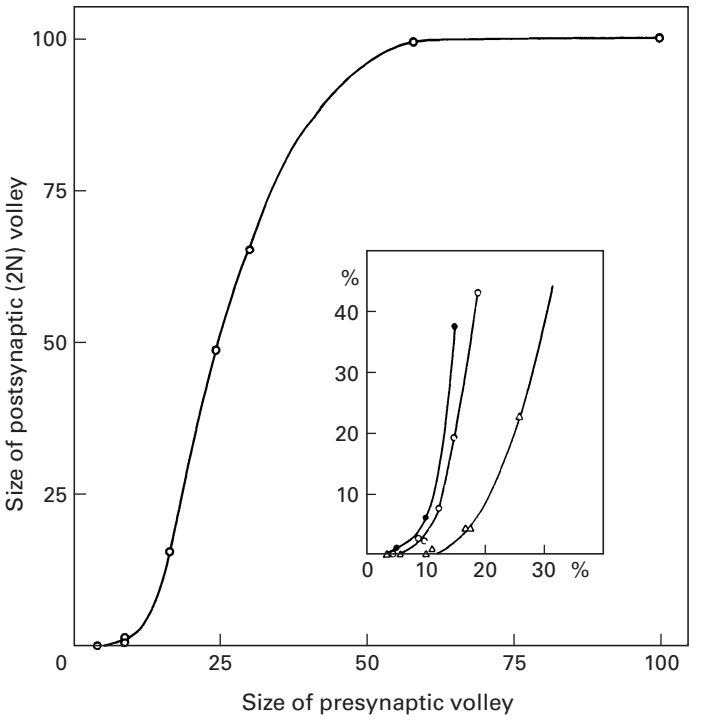


Figure 6.2
Input-output curve representing monosynaptic reflex amplitude in response to the size of the afferent volley. The inset shows that in multiple experiments there was a clear threshold of afferent input, between 5 and 10%, that formed a threshold for the population response. Reproduced from Lloyd [Llo43] with permission.

Single populations give sigmoidal activation curves (figure 6.4A). Multiple populations destroy the simple sigmoid shape, but reflect the multiple inflection points generated by the different population contributions (figure 6.4B).

If one assumes that the level of excitation of a cell decays as $\alpha(t)$, then the average level of exciation generated in an E cell at time t is

$$x(t) = \int_{-\infty}^{t} \alpha(t - t')[w_1 E(t') - w_2 I(t') + P(t')]dt' \tag{6.3}$$

where w_i is the average number of synapses on each cell, and $P(t)$ is the external input to the E population ($Q(t)$ will be the input to the I population). The activity in a subpopulation at time $t + \tau$ will then be the product of the cells at time t that are both sensitive and above threshold. Given the proportion of cells not refractory in (6.1), the value of $S(x)$ in (6.2), and the formula for $x(t)$ in (6.3), Wilson and Cowan wrote

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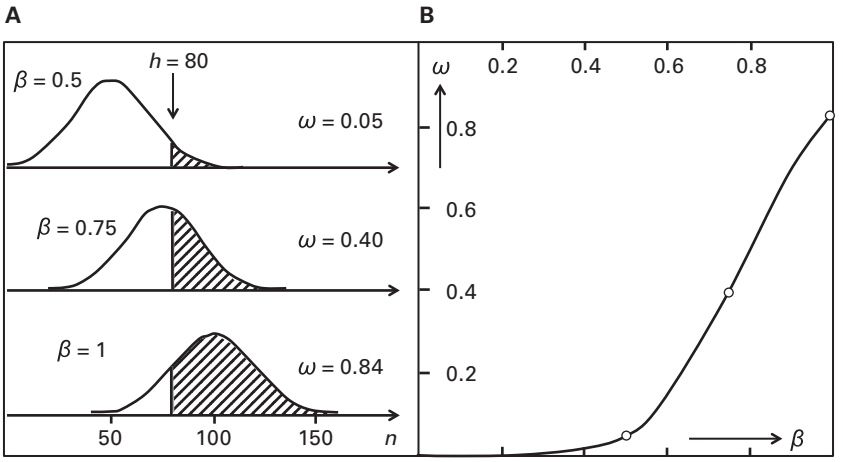


Figure 6.3
Demonstration of how the fractional activation, β, of Gaussian activation curves of afferent neurons, and threshold, h, shown in A, would integrate the cumulative activation to produce sigmoidal output functions shown in B. Modified from [Ral55b] with permission.

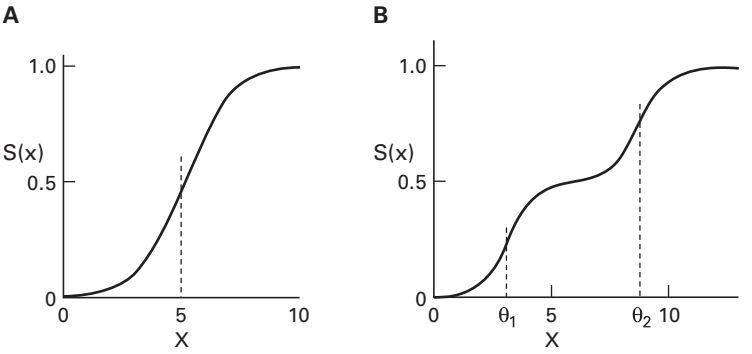


Figure 6.4
Population activation curves from unimodal, (A) and multimodal population input (B). Multiple modes can be produced by multiple distributions of thresholds, or afferent synapses. Reproduced from [WC72] with permission.

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$$E(t+\tau) = \left[1 - \int_{t-r}^{t} E(t')dt'\right] \cdot S_e \left\{\int_{-\infty}^{t} \alpha(t-t')[w_1 E(t') - w_2 I(t') + P(t')]dt'\right\}$$
$$I(t+\tau) = \left[1 - \int_{t-r}^{t} I(t')dt'\right] \cdot S_i \left\{\int_{-\infty}^{t} \alpha(t-t')[w_3 E(t') - w_4 I(t') + Q(t')]dt'\right\} \tag{6.4}$$

Temporally coarse graining these functions E and I so that they are replaced by a moving time average for a small interval will remove rapid temporal fluctuations that are not essential for the overall population dynamics that they sought

$$\int_{t-r}^{t} E(t')dt' \rightarrow \bar{E}(t)t'\Big|_{t-r}^{t} = r\bar{E}(t)$$

$$\int_{-\infty}^{t} \alpha(t-t')E(t')dt' \rightarrow \bar{E}(t)t'\Big|_{t-k}^{t} = k\bar{E}(t)$$

where the r and k reflect the times that the refractory period and synaptic activation lingered, respectively. Taylor expanding about $\tau = 0$ for $E(t+\tau)$ and $I(t+\tau)$

$$E(t+\tau) = E(t) + \frac{dE}{dt}(t+\tau-t) + \ldots = E(t) + \tau\frac{dE}{dt}$$

$$I(t+\tau) = I(t) + \frac{dI}{dt}(t+\tau-t) + \ldots = I(t) + \tau\frac{dI}{dt}$$

leads to

$$\tau\frac{d\bar{E}}{dt} = -\bar{E}(t) + (1 - r\bar{E})S_e\left\{k[w_1\bar{E}(t) - w_2\bar{I}(t) + P(t)]\right\}$$
$$\tau'\frac{d\bar{I}}{dt} = -\bar{I}(t) + (1 - r\bar{I})S_e\left\{k'[w_3\bar{E}(t) - w_4\bar{I}(t) + Q(t)]\right\} \tag{6.5}$$

Has all of this slogging through this original derivation been worthwhile to the reader? I know of no other way to, for instance, have any idea why in the Wilson-Cowan equations as they are now used (see, e.g., equations (6.7)) there appear equivalent terms to $-\bar{E}(t)$ and $-\bar{I}(t)$ as shown in (6.5). Some things cannot be hand-waived away.

Is the coarse graining used in (6.5) reasonable? Wilson and Cowan compared the results in figure 6.5. They argued that the fast time scale oscillations from (6.4) are not important.

Finally, they proposed that the rest state was stable to small perturbations. So they shifted their sigmoid activation functions so that $E = 0$ and $I = 0$ when $S_e(0) = 0$ and $S_i(0)$, respectively. We now know that the so-called rest state of cortex is highly active [Len03, DRFS01]. Nevertheless, we also suspect that such states still have some structural

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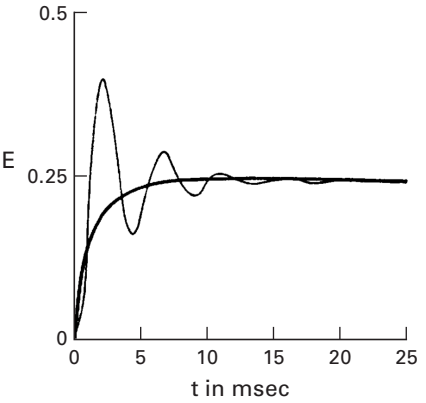


Figure 6.5
Comparison of (6.4) (light line) with (6.5) (heavy line). Reproduced from [WC72] with permission.

stability to perturbations (a more detailed consideration of such perturbations can be found in [UCJBS09]).

These equations could have nullclines plotted on a phase plane, illustrating the case that the excitatory (P) and inhibitory (Q) inputs are zero. Two stable and one unstable equilibrium are shown in figure 6.6.

The effect of positive P or Q is to shift these nullclines and alter the stability of these fixed points. This generates changes in the qualitative nature of the cortical dynamics modeled. When Wilson and Cowan varied the excitatory population excitation P up and down, they noted hysteresis about two stable states in their solutions, as shown in figure 6.7A. A considerable amount of relative excitation P was required to trigger the high-activity state—a *population threshold* that was an emergent property of these equations.

The results, in figure 6.7A, were not dependent on inhibition being present. The more complex multiple hysteresis loops shown in panel B depend on the presence of inhibition. Hysteresis is one way to envision modeling short-term memory in such models. Wilson and Cowan were aware of hysteresis effects seen in visual cortex experiments when they modeled this result.

The next phenomenon that they observed in this model was that the duration of inputs affected the population response, as shown in figure 6.8A. In fact, such a strength-duration curve is very similar to that of accommodating single neurons, including an effective rheobase for the population.

Other important phenomena that the model demonstrated of physiological significance are damped oscillations in response to brief stimulation, similar to those of evoked potentials recorded from cortex (figure 6.8B).

Wilson and Cowan found a range of excitatory input P for which the nullclines would produce an unstable fixed point, about which would emerge a stable limit cycle, as shown in figure 6.9.

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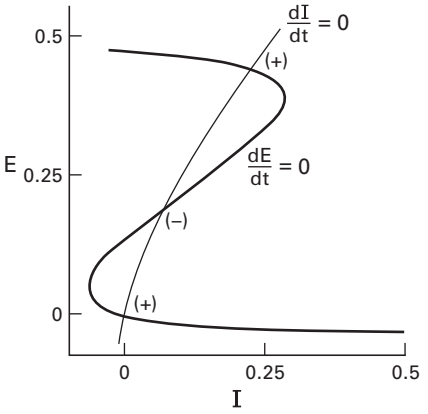


Figure 6.6
Phase plane representation of Wilson-Cowan equations showing nullclines with two stable (+) and one unstable (−) fixed points for the case that the excitatory (P) and inhibitory (Q) inputs are zero. Reproduced from [WC72] with permission.

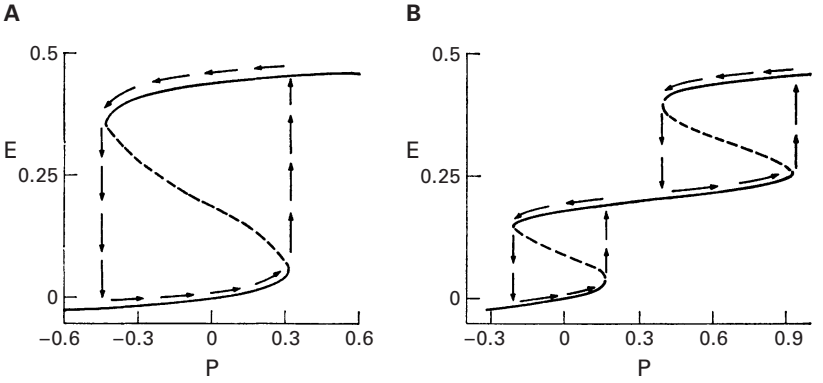


Figure 6.7
Hysteresis observed in excitatory population active, E, in relation to excitatory activation, P, shown in A. Multiple hysteresis loops in B require inhibition. Solid lines are stable states, whereas dashed lines are unstable states. Reproduced from [WC72] with permission.

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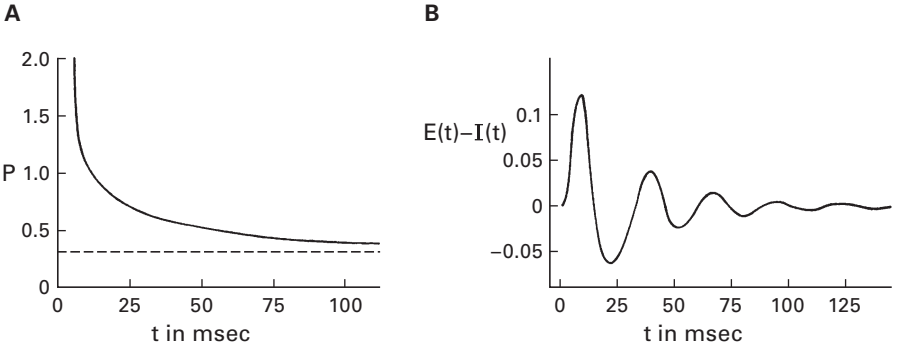


Figure 6.8
Excitatory current strength, P, versus duration required to activate cortex (A), so as to excite it to a high level of activity as in figure 6.7. B shows the ability of the model to demonstrate damped oscillations. Reproduced from [WC72] with permission.

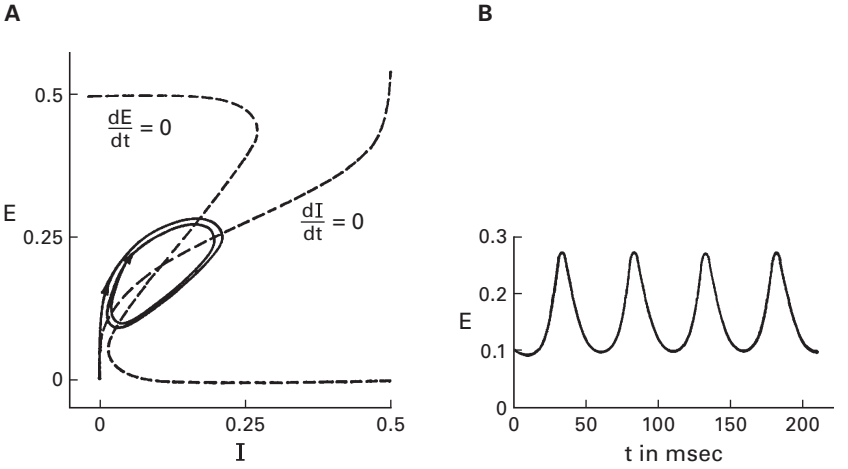


Figure 6.9
Emergence of stable limit cycle about unstable fixed point in A, and time series of E shown in B. Reproduced from [WC72] with permission.

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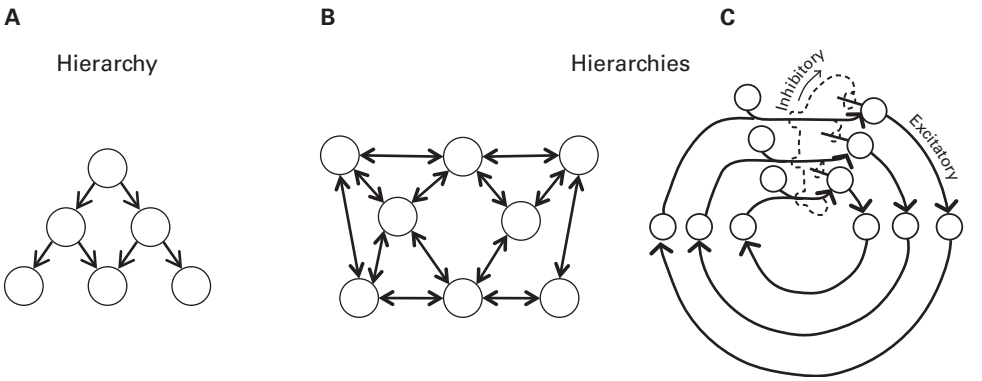


Figure 6.10
Contrast between hierarchy (A) and two different types of heterarchy, the fishnet network (B) and McCulloch's circular heterarchy, shown in C, modified from [McC45] with permission.

The implication of limit cycles for cortex was that their amplitude and frequency could encode stimulus intensity. Furthermore, limit cycle stability implied noise insensitivity to such encoding. All of their dynamical findings, Wilson and Cowan stress, are consequences of the nature of the nullclines in their phase space representation. The precise form of the model and value of parameters are of secondary importance.

6.3 Wilson and Cowan during 1973

In the previous section, we studied a general model of cortical subpopulation interactions that had no spatial extent. The analogy is somewhat like the Hodgkin-Huxley equations of the space-clamped membrane patch (chapter 3). The Hodgkin-Huxley equations needed to be extended into the one-dimensional propagating action potential. For the dimensionless Wilson-Cowan equations, we need to extend them to the two-dimensional cortex.

An important concept that Wilson and Cowan introduced in their 1973 paper [WC73] was that cortex was more likely organized as a *heterarchy*, rather than a *hierarchy*. The concept of heterachy is generally attributed to Warren McCulloch in his 1945 paper, and the difference between hierarchies and heterarchies is schematized in figure 6.10.[5]

In their 1973 paper, Wilson and Cowan developed a mathematical model that represented cortex as a complex, interconnected heterarchy of quasi-two-dimensional sheets. Neurons were assumed to be uniformly distributed within the sheet. They assumed that lateral connectivity was a function of distance only. Furthermore, they assumed that the sheet was

5. McCulloch's papers are famously difficult to read. An interesting philosophical discussion can be found in the Web-published paper by von Goldammer and colleagues [VGPN03].

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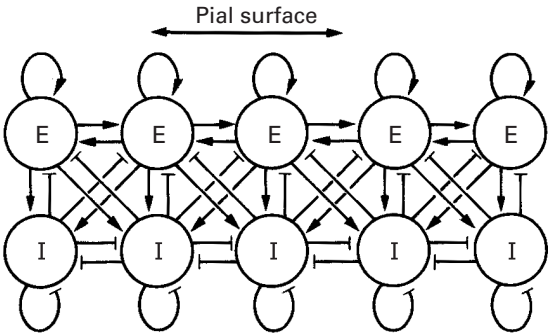


Figure 6.11
Schematic of general two-dimensional model of cortex. The aggregates of E and I cells are shown with all possible interconnectivities. Reproduced from [WC73] with permission.

isotropic (there was no direction preference). Again as in 1972, they assumed that there were two types of neurons—excitatory and inhibitory—and that there were equal numbers of each.[6] Importantly, they presumed that reliability from unreliable components would be achieved by redundancy.

The schematic of their 1973 model is shown in figure 6.11. In the figure, the excitatory, E, and inhibitory, I, nodes are not representative of single cells, but rather spatially localized aggregates of such cells. All possible interconnectivities are represented, but it is assumed that the probability of connection falls off exponentially with distance. Some rather severe assumptions enter into this model. They assume that connectivity is similar between all neurons. They assume that the conduction velocities were effectively infinite—in other words much faster than the dynamics of cortical patterns that emerge from this model.

The evolution of the equations from the 1972 paper,

$$\tau \frac{d\bar{E}}{dt} = -\bar{E}(t) + (1 - r\bar{E})S_e\left\{k[w_1\bar{E}(t) - w_2\bar{I}(t) + P(t)]\right\}$$

$$\tau' \frac{d\bar{I}}{dt} = -\bar{I}(t) + (1 - r\bar{I})S_e\left\{k'[w_3\bar{E}(t) - w_4\bar{I}(t) + Q(t)]\right\}$$

to the 1973 paper,

$$\begin{aligned}
\tau \frac{\partial \bar{E}(x,t)}{\partial t} &= -\bar{E}(x,t) + (1 - r_e\bar{E}(x,t))S_e\left\{k[w_{ee} \otimes \bar{E}(t) - w_{ie} \otimes \bar{I}(t) + P(t)]\right\} \\
\tau' \frac{\partial \bar{I}(x,t)}{\partial t} &= -\bar{I}(x,t) + (1 - r_i\bar{I}(x,t))S_i\left\{k'[w_{ei} \otimes \bar{E}(t) - w_{ii} \otimes \bar{I}(t) + Q(t)]\right\}
\end{aligned} \tag{6.6}$$

6. Equal numbers of excitatory and inhibitory neurons is not a very accurate assumption, as we know today, but probably not a major problem if the parameters of the model are properly adjusted.

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are shown here, where $\otimes$ indicates convolution. In regions of densely packed neurons, such as within cortical columns, the convolutions can be replaced by products of appropriate averages.

Wilson and Cowan make three important assumptions on their parameters: (1) the rest state is stable to small perturbations, (2) there is no uniformly excited state in the absence of a maintained stimulus (otherwise it would be in a seizure-like state), and 3) the inhibitory connections are longer range than the excitatory connections.

Assumption 3 is critical to form inhomogenous patterns in excitable media. The underlying physics were laid down by Alan Turing in 1952 for the chemical basis of morphogenesis. Ordinarily diffusion is stabilizing—it smooths things out. But for multiple chemical species, this requires the speed of diffusion of the species to be similar. In chemical reaction-diffusion systems with fast- and slow-diffusing reactants, an instability can be created whereby inhomogenous patterns may form. Neural networks formally share some of the same physics as these chemical systems, in that the fast-diffusing reactant may be likened to the further connectivity reach of the inhibitory neurons.[7]

These Wilson-Cowan equations produced a wide variety of dynamical modes, which included active transients, spatially localized limit cycles, and spatially inhomogenous stable steady states.[8] In the spirit of Hodgkin and Huxley, Wilson and Cowan extensively show that their model can account for a wide variety of neuronal phenomena that had been previously experimentally observed. In contrast to Hodgkin and Huxley, they started from general principles, and then sought what phenomena they could simulate. Hodgkin and Huxley started from experimentally derived equations.

Both temporal summation (figure 6.12A) and spatial summation (6.12B) could be seen in the threshold to fire self-regenerative active transients. Such bumps of activity are analogous to single-action potential spikes in Hodgkin-Huxley or Fitzhugh-Nagumo equations.

Edge enhancement could be observed. This phenomena was increased as the strength of stimuli were increased, as shown in figure 6.13.

Spatially localized limit cycles, in response to constant but spatially localized stimuli, are shown in figure 6.14. The frequency of the limit cycles encoded stimulus strength.

They noted frequency demultiplication (recall Van der Pol 1926 [VdP26], chapter 4), which had by then been observed in driving the thalamus, and illustrated in figure 6.15.

For sufficiently *large* values of inhibitory activation, Q, they observed traveling waves, as shown in figure 6.16. Such traveling waves will be essential in our modeling of cortical waves later in this chapter. Intriguingly, Wilson and Cowan predicted propagation velocities

7. It is hard to improve on the reaction-diffusion analogy of grasshoppers and forest fires in James Murray's fine book on *Mathematical Biology* [Mur03].

8. A superb and authoritative description of the dynamics of these equations, can be found in Hugh Wilson's book *Spikes, Decisions, and Actions*, accompanied by computer code carrying out many of the difficult computations using the original formulation of the equations [Wil99]. A concise and deep mathematical perspective on these equations can be found in Ermentrout and Terman's recent book [ET10].

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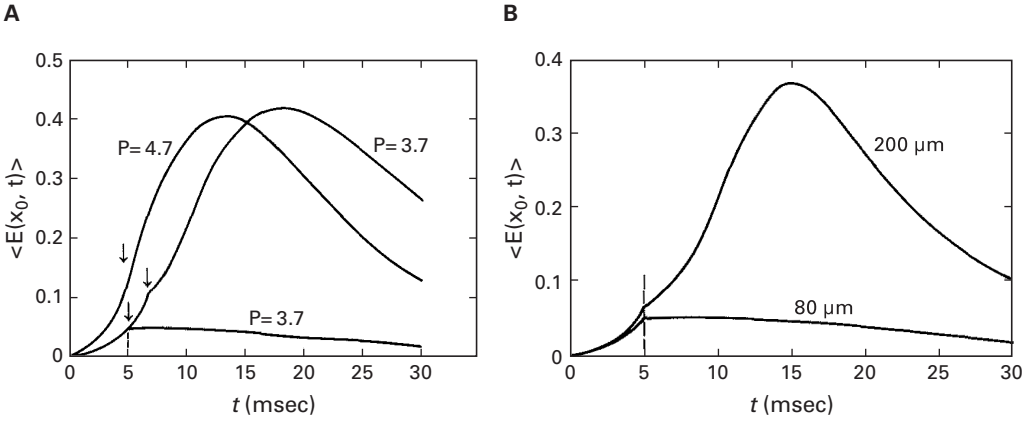


Figure 6.12
A shows temporal summation. At the same stimulus intensity, the longer stimulation time leads to an active transient. In B, the larger spatial stimulus application with the same current intensity per node leads to an active transient. Modified from [WC73] with permission.

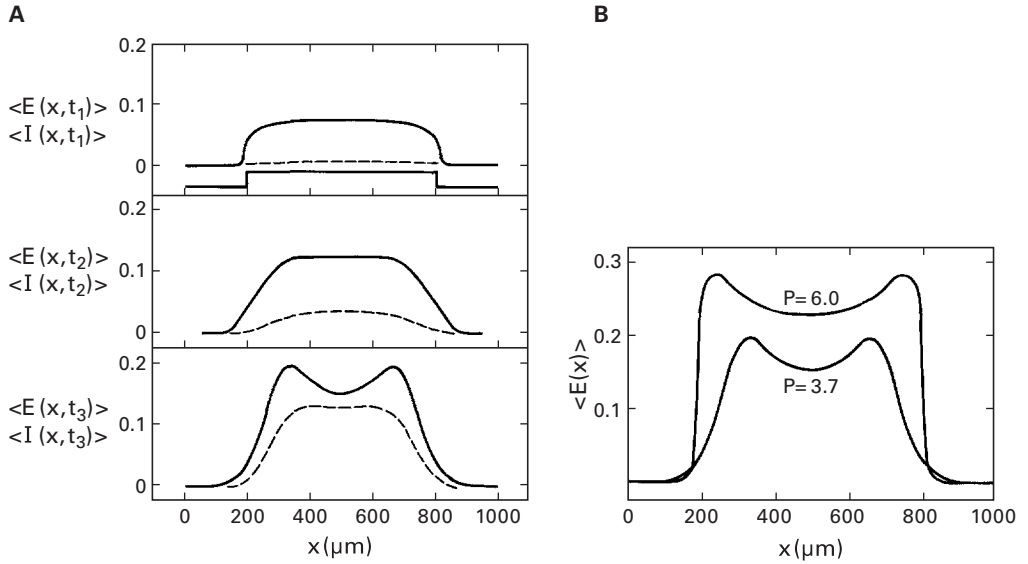


Figure 6.13
Example of edge enhancement shown in the temporal profiles of excitation (solid) and inhibitory (dashed) lines in A, and increasing spatial disparity in the maximal edge enhancement is brought out by increasing stimulus strength in B. Modified from [WC73] with permission.

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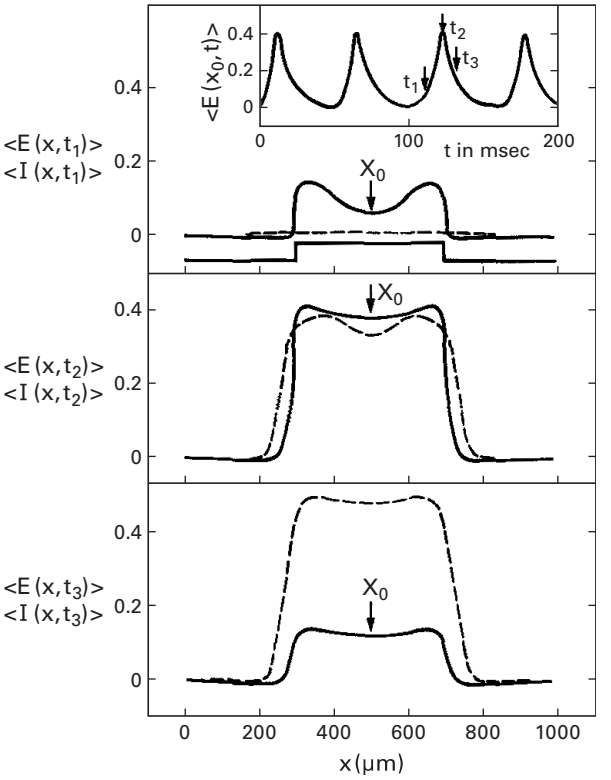


Figure 6.14
Spatially localized limit cycles. Reproduced from [WC73] with permission.

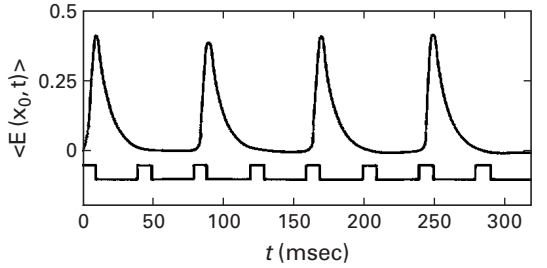


Figure 6.15
Example of frequency demultiplication. Reproduced from [WC73] with permission.

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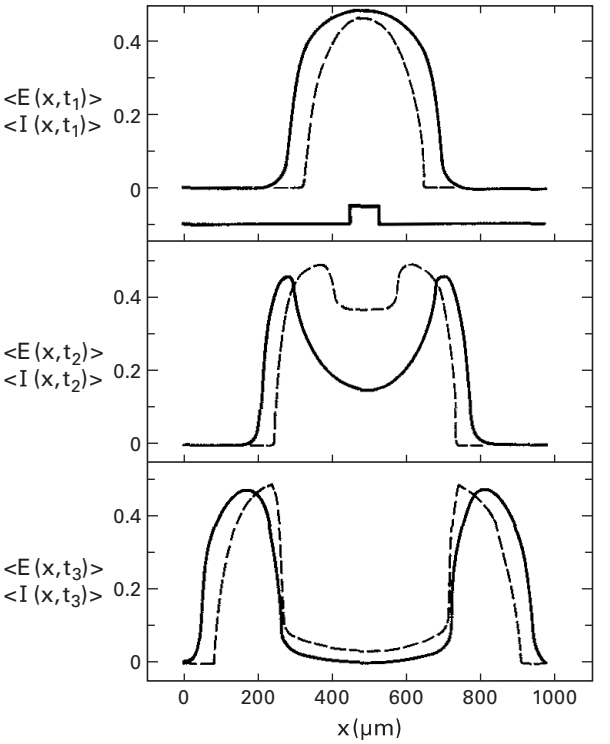


Figure 6.16
Traveling waves in Wilson-Cowan equations. Reproduced from [WC73] with permission.

of such traveling waves in the neighborhood of 4 cm/sec, exactly what we observe now with cortical traveling waves [RSG05].

Finally, spatially inhomogeneous localized activity could be observed as shown in figure 6.17. Such stationary bumps are what would be, if the individual neuron activities were observed, reverberant firing activity. Wilson and Cowan likened this to the neural correlate of working memory observed by Fuster and Alexander [FA71].

6.4 Wilson and Cowan after 1973

In 2001, David Pinto and G. Bard Ermentrout modified the Wilson-Cowan equations [PE01a, PE01b]. First, in the region of sparse activity, that is when $r_{e/i}$ is small, $(1 - r_e \bar{E}(x, t)) \to 1$ in equations (6.6), and similarly for the $\bar{I}$ equation.[9] For potentially disinhibited cortex,

9. This is a reasonable assumption from what we are learning about the actual sparseness of neuronal activity [Len03].

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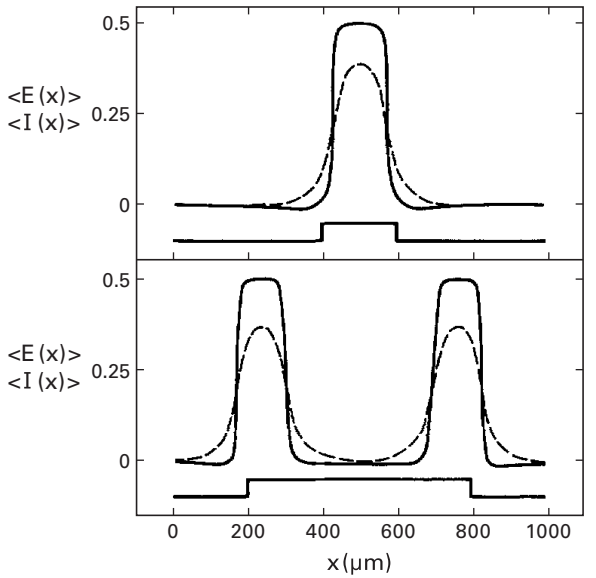


Figure 6.17
Spatially inhomogeneous localized activity. Reproduced from [WC73] with permission.

as they were seeking to model, they added a slow recovery variable to just the excitation dynamics as (switching notation a bit)

$$\dot{u}(x,t) = -u(x,t) + \int_{-\infty}^{\infty} w(x-x')P(u(x',t)-\theta)dx' - a(x,t)$$
$$\frac{1}{\varepsilon}\dot{a}(x,t) = -\beta a(x,t) + u(x,t) \tag{6.7}$$

Here, u is activity (equivalent to Wilson and Cowan's $\bar{E}$), a is the slow recovery variable, θ is threshold, P is a synaptic firing rate relative to the threshold, and $\epsilon < 1$ slowing down $\dot{a}$ relative to $\dot{u}$. Note the $-u(x,t)$ term is the same as the $-\bar{E}$ term in equation (6.5). Variable a is equivalent to the action of both potassium, as well as inactivation dynamics and spike frequency adaptation. But keep in mind that we are in the Wilson-Cowan world—there are no individual neurons, only neuronal aggregates and local averages of dynamics. This formulation will be particularly useful if one were to perform experiments where traveling waves would be incited in cortex under conditions where inhibition was pharmacologically blocked [PE01a, HTY+04]. For the same reason that it was useful to Pinto and Ermentrout, we will shortly apply this version of the following Wilson-Cowan equations to cortical wave experiments.

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In a second paper [PE01b], Pinto and Ermentrout extended this model to include inhibition. The most general way of doing this would be to have two sets of two-dimensional lattices, formulated as

$$\begin{aligned}
\dot{u}(x,t) &= -u(x,t) + \int_{-\infty}^{\infty} w_{ee}(x-x')P(u(x',t)-\theta)dx' - a(x,t)\cdots \\
&\quad - \int_{-\infty}^{\infty} w_{ie}(x-x')P(v(x',t)-\theta)dx' \\
\frac{1}{\varepsilon}\dot{a}(x,t) &= -\beta a(x,t) + u(x,t) \\
\dot{v}(x,t) &= -v(x,t) + \int_{-\infty}^{\infty} w_{ei}(x-x')P(u(x',t)-\theta)dx' - b(x,t)\cdots \\
&\quad - \int_{-\infty}^{\infty} w_{ii}(x-x')P(v(x',t)-\theta)dx' \\
\frac{1}{\delta}\dot{b}(x,t) &= -\lambda b(x,t) + v(x,t)
\end{aligned} \tag{6.8}$$

where u represents excitatory and v inhibitory neurons, and w_{ee}, w_{ie}, w_{ei}, w_{ii}, the connectivity footprints from $e \to e$, $i \to e$, $e \to i$, and $i \to i$, respectively. I have written two recovery variables, a and b, with potentially separate recovery time constants $1/\varepsilon$ and $1/\delta$.

If you are modeling full brain dynamics, such as EEG from epilepsy patients, then you need to start from this more general formulation for excitatory and inhibitory interactions in equations (6.8). Not all of the terms may be necessary ([PE01b] often eliminated w_{ii}, for instance). And the symmetry considerations, to be discussed later, are also critical to take into account. If, however, you are going to start with cortex creating oscillatory waves when inhibition is blocked, then the formulation in equations (6.7) will serve us well for the remainder of this chapter.

6.5 Spirals, Rings, and Chaotic Waves in Brain

My colleague William Troy has long been interested in oscilatory and wave phenomena. When he began to be interested in models of neural cortex [LTGE02, LT03], the complex bumps and spiral waves were intriguing. After the work of Beurle mentioned earlier, there were demonstrations in homogenous integrate and fire networks that target waves evolving

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toward spiral waves could be observed, often following periodic driving of the network (see [MC93] and note references within to previous work in inhomogenous excitable networks). But it was not at all apparent that these phenomena were seen in the brain to the degree that they were seen in mathematics. It is not that it was hard to produce waves of activity in neural tissue, but the optical techniques capable of replacing traditional electrode measurements to image complex waves were relatively recent [BW03]. The problem was that most brain slice preparations were done coronally or axially [TC98], and because of the layered strucure, waves were not produced in a homogenous two-dimensional media as in Troy's mathematics. From a math meeting we placed a phone call to an experimentalist that we knew was doing experiments in a different type of brain slice and interested in wave physiology—Jian-young Wu.

Wu was using special tangential slices that were not commonly studied physiologically. Several years before, Michael Gutnick and colleagues [FBG98] had used such tangential slices in their studies of layer 4 in barrel cortex physiology (figure 6.18A), and had noted that in the presence of pharmacological disinhibition with bicuculline, all-or-none traveling waves could be initiated (figure 6.18B). In distinction to [FBG98], Wu removed the superficial layers of cortex, and then cut a tangential slice from visual cortex (figure 6.18C) that contained layers 3, 4, and 5 (of the six-layer cortex). These layers contained some of the largest neurons, and wavelike phenomena were known to occur preferentially in these layers and likely were instigated in layer 5 in in vitro studies [TC98]. We asked him if he could try to find spiral waves. It was not many weeks before he, and his postdoc Xiaoying Huang, had found many.

Spiral waves are generic in physical, chemical, and mathematical studies of two-dimensional excitable media. Such structures emerge in the presence of isotropy, where there are no preferred directions in the two-dimensional plane. It is the anisotropy that makes coronal or axial slice cuts problematic for correlating brain activity waves with the abundance of physical and mathematical theory available. The rodent visual cortex has no ocular dominance columns [GSL99], and is a particularly attractive cortex if a homogenous and isotropic one is sought.[10]

Rotational waves had been observed before in turtle brain experiments [PCP+97]. But they were difficult to see in the intact cortex, and the signal processing required to extract them from optical imaging data were cumbersome.

These new waves in mammalian tangential cortical slices were robust [HTY+04]. They were observed using voltage-sensitive dye. Such measurement technique gives a local mean field average of transmembrane potential through a full-thickness, small cylindrical core (about 100×100 μm) of the piece of brain under each photodiode (figure 6.18D). The full photodiode array imaged 128 or, later, 472 pixels. The data were exactly what the Wilson-Cowan equations, or the Pinto-Ermentout variant, was designed to model.

10. Whether the barrel cortex gives the same wave properties as visual cortex, or might impose more inhomogeneity to wave propagation much like bumper pool from the barrels, remains an open question.

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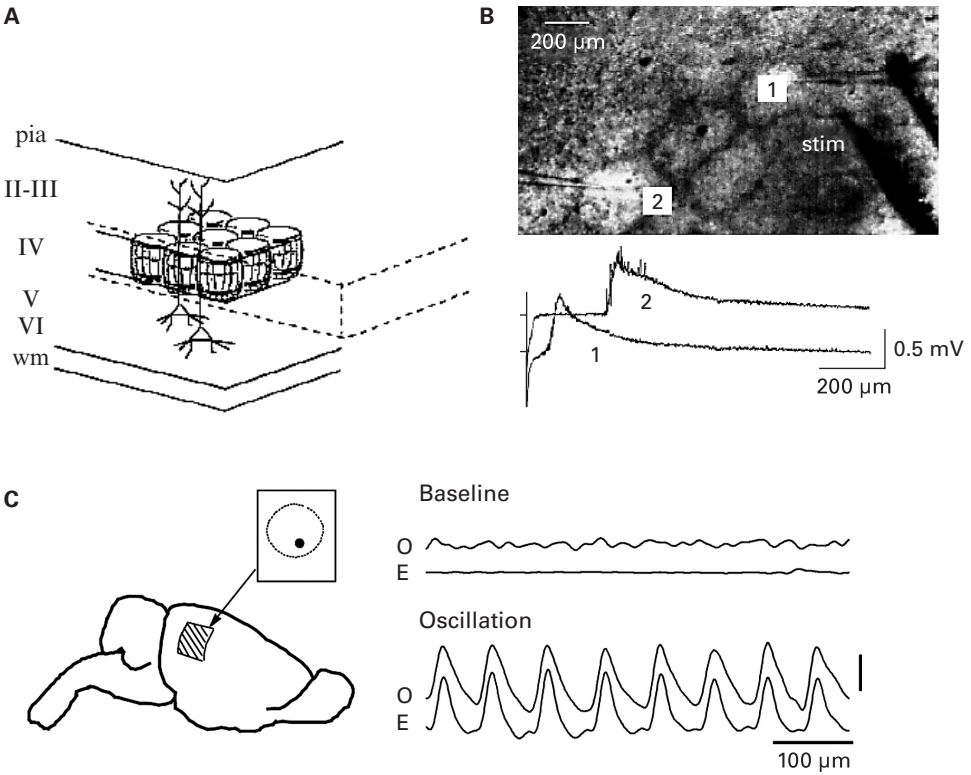


Figure 6.18
A shows layer-specific tangential slice from barrel cortex, and B shows all-or-none traveling waves. A and B reproduced from [FBG98] with permission. C shows the tangential slices from visual cortex, with oscillations that are similar whether electrically (E) or optically (O) recorded. Reproduced from [HTY+04] with permission.

Brain slices have most of their connections to the rest of the brain cut, and one always seems to need to add something to activate them in order to restore a level of network activity. Wu and colleagues used carbachol, which activates acetylcholine receptors, and bicuculline, which blocks inhibitory gamma-aminobutyric acid type-a (GABAa) channels that are fast inhibitory synaptic receptors. Exposed to these compounds, robust oscillations with organized wave activity could be readily seen in figure 6.19 (plate 5). These oscillatory episodes could be seen spontaneously, but they were typically triggered with small stimuli for aligning them with the optical recordings and to minimize photobleaching effects. Just like action potentials, they had a refractory period afterwards, within which new episodes could not be triggered with stimuli. A variety of events are seen in vertebrate nervous systems such as up states [SHM03], spinal cord burst firing [CMO06], animal [ZCBS06] and human [SSKW05] epileptic seizures, and these cortical wavelike events [HTY+04]

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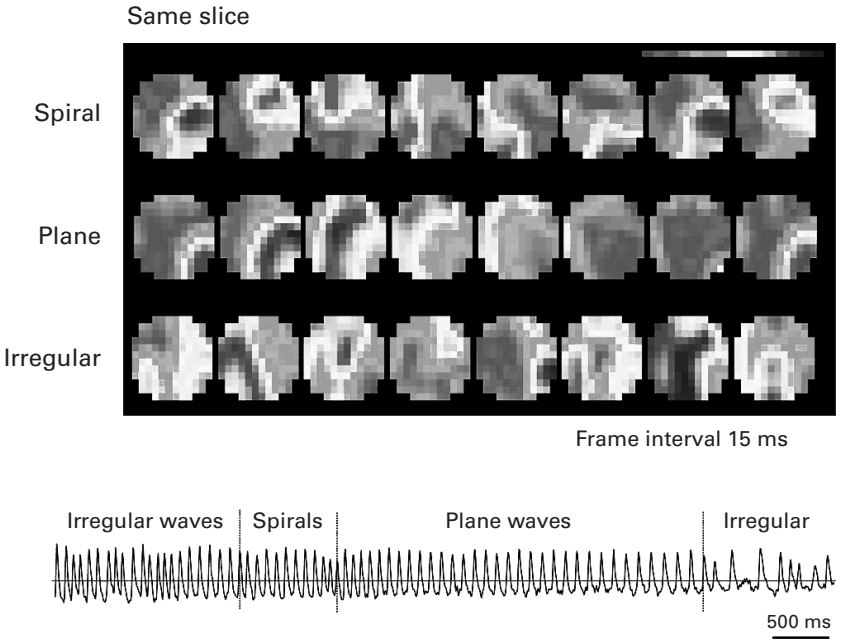


Figure 6.19 (plate 5)
From a single oscillatory episode, a progression is seen of chaotic irregular, to spiral, to plane, and back to chaotic waves. Reproduced from [HTY+04] with permission.

share the properties of being episodic, oscillatory, and having apparent refractory periods, following which small stimuli can both start and stop such events.

Interestingly, we saw no clear anatomic interference with the waves that formed. Although we understood that some degree of irregularity or symmetry breaking was likely to nucleate a wave, the spiral centers wandered about the two-dimensional field when we observed them. Sometimes a spiral would rotate clockwise, and the next time one formed, it would rotate counterclockwise. There appeared to be a true singularity of phase—one pixel could be found that was surrounded by the greatest spatial phase gradient. From a mathematical point of view, our observations were consistent with a degree of translation invariance that would permit such waves to be analyzed with many of the tools used to study other excitable physical systems.

We collaborated with Carlo Laing and William Troy to attempt to simulate such waves using the Wilson-Cowan equations.[11] They employed a partial differential equation solution to solve these equations, and produced qualitatively similar ring, plane, and spiral wave

11. I suppose it would be appropriate were I to designate the form of these equations as the Pinto-Ermentrout variant of the Wilson-Cowan equations. But for the rest of this text, I will lump and designate all variants and derivatives of the Wilson-Cowan equations as Wilson-Cowan.

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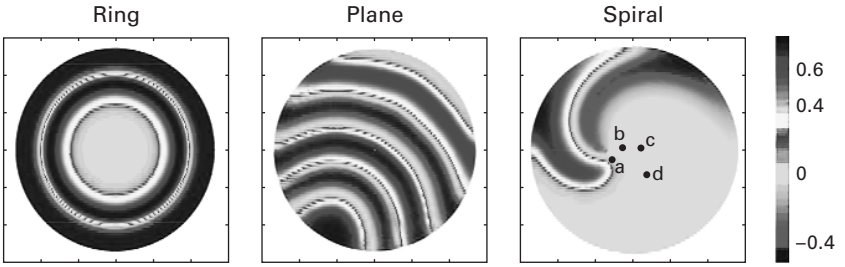


Figure 6.20 (plate 6)
Simulation of ring, plane, and spiral waves with the Wilson-Cowan equations. Reproduced from [HTY+04] with permission.

patterns as were observed experimentally (figure 6.20, plate 6). Simulation is not proof of mechanism, but these calculations show that the rather simple generic principles of cortical organization within the Wilson-Cowan equations are capable of emulating a wide range of the dynamics we observed in these experiments. What we did not simulate at this time were the complexities of the irregular and chaotic spatiotemporal patterns that often initiated and resolved such oscillatory episodes. One can, indeed, see much more irregular patterns within the Wilson-Cowan system, but it requires some careful setting of the initial conditions prior to integration. A full exploration of the basin of attraction of these equations has not, to the author's knowledge, been done. But the other feature prominent in the experiments that is left out of the simulation is that these wave episodes *evolve* with time. The Wilson-Cowan equations are deterministic, and with a given set of initial conditions, will (after transients resolve) yield one dynamical pattern. Real slices, and real brains, never settle down to just one stable pattern. We will cover this subject later in considerably more detail (chapter 7).

Intriguingly, the difficulties in earlier experiments in observing waves of activity in brains [PCP+97] could be overcome in a variety of ways. One is to employ optical imaging such as just discussed, using slices of brain to focus on relevant regions in the tissue. In recent years, waves of activity in intact brains in the visual cortex have been observed optically [XHTW07]. An alternative is to employ arrays of sharp electrodes in intact cortex, using the electrodes to penetrate down to the layers exposed here in slices. A recent study of waves in the motor cortex employed such penetrating electrodes to considerable success [RRH06], as did a recent study of hippocampal traveling waves during navigation [LS09].

In terms of modeling and tracking, it is important to perhaps reiterate that our goal in this book is *not* to reengineer the brain. We don't particularly care if the Wilson-Cowan equations actually reflect the detailed dynamical mechanism going on within our observed cortex. We want to find efficient models that have the dynamical repertoire of the brain. We want them to be only as complex as needed to perform such emulation. This latter need is so that the parameters can be most accurately determined, so that the tracking is most accurate. Employing an accurate but hugely complex model, even though it might be a true reflection of the brain dynamics, is not going to serve us well if the parameter and

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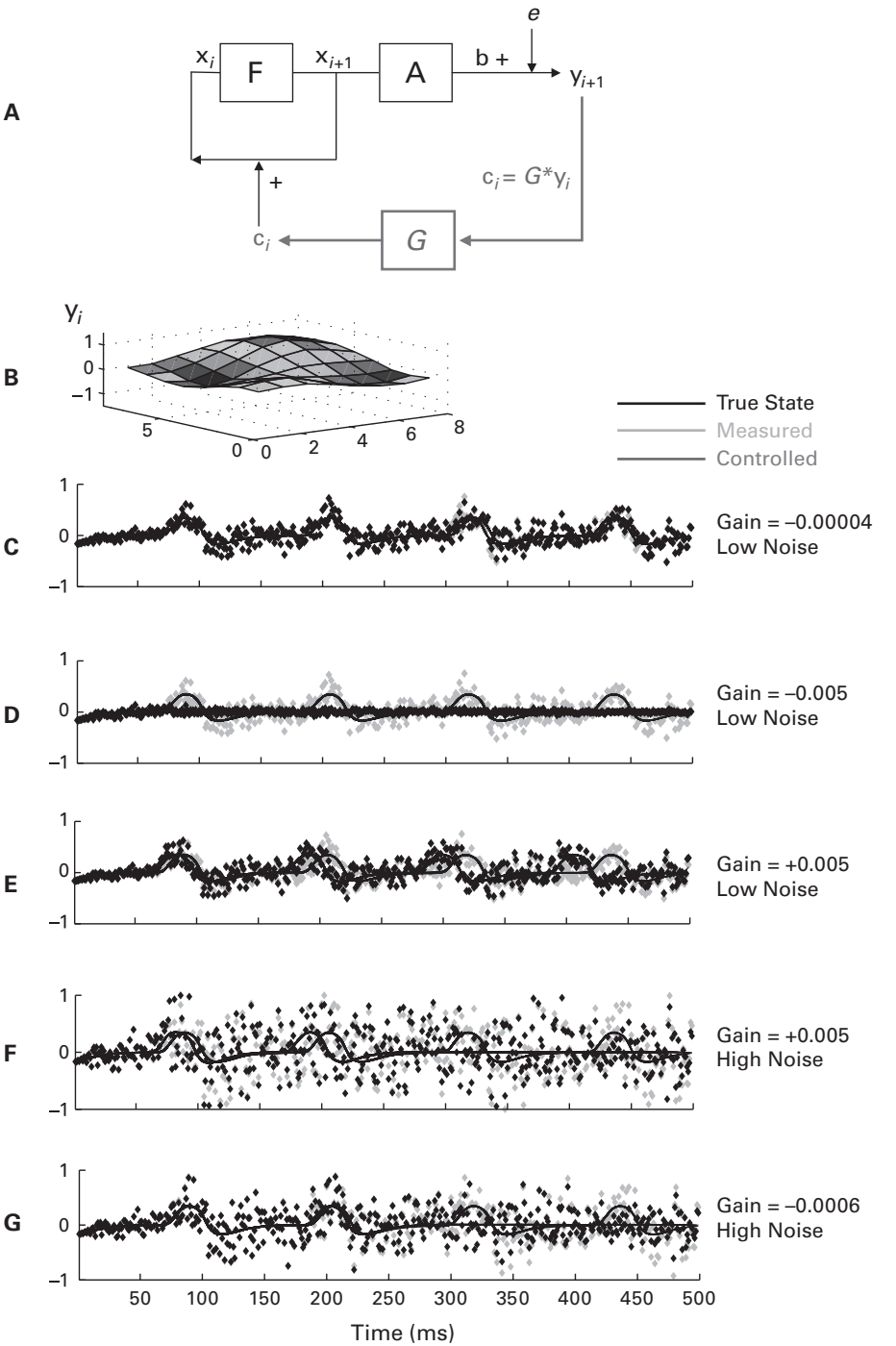


Figure 6.21 (plate 7)
(Caption on facing page)

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variable fits are so poor that the model cannot track the observed dynamics well. Last, we need prediction—short-term prediction of the brain dynamics. We assume that a model that is fundamental and based on our growing knowledge of neuroscience will, at some point, perform better than an empirical statistical model. And given all of the above, the Wilson-Cowan model is perfect for our next efforts at using them to track and control brain dynamics.

6.6 Wilson-Cowan in a Control Framework

We now take the Wilson-Cowan formalism and wrap it within a control formalism.

We begin with an 8×8 grid of discrete Wilson-Cowan equations as in [SS08]

$$\begin{aligned} \dot{u}(x, y, t) &= -Cu(x, y, t) - a(x, y, t) + K \sum_{\substack{i,j \\ i \neq j}} w(x, y, p_i, q_j) H(u(p_i, q_j, t) - \theta) \\ \tau \dot{a}(x, y, t) &= Bu(x, y, t) - a(x, y, t) \end{aligned} \tag{6.9}$$

where

$$w(x, y, p, q) = e^{-k\left((x-q)^2+(y-p)^2\right)}$$

In reading this notation, (x, y) represents the location that the function is evaluated at, and (p, q) are dummy variables that represent all other positions in the two-dimensional plane that are examined for whether the activity, u, is at least as large as threshold θ. The footprint function $w(x, y, p, q)$ falls off as a symmetrically decreasing Gaussian function with distance. H is the Heaviside function, equal to 1 when the activity at $u_{i,j} \geq \theta$, and equal to 0 when $u_{i,j} < \theta$, where θ is the threshold. τ is the differential time constant between the two equations—the rate of change of a is slower than the rate of change of u. C, K, B, and k are adjustable parameters.

In figure 6.21A (plate 7), we show the schematic for the Wilson-Cowan equations represented as the dynamical process F. We observe the dynamics through the process A, which

Figure 6.21 (plate 7)
(A) Schematic of dynamical system F, which acts on state variable vector x, and observation function A, which produces output vector b, to which measurement noise e is added, generating output observable y. When direct proportional control is used, control function G is employed, which generates the control vector c. This control vector is directly "injected" into the neuronal elements, adding vector c to the activity variable u (nothing is added to the unobserved variable a or to the parameters of the system). (B) An instantaneous snapshot of the 8×8 output y during a rotational spiral. (C) The output of the observable y under conditions of small negative gain and low noise, from a single element of the grid in B, shown as a time series without (blue) and with (red) control applied. The underlying state vector activity element u_i is plotted as a thin black line. In contrast with C, D shows outputs under conditions of higher negative gain and low noise, E shows positive gain and low noise, and F and G show high noise conditions with positive and negative gains, respectively. Low noise corresponds to 1 to 5, and high noise corresponds to 10 to 14 on an arbitrary scale. Note that under high noise conditions, the application of direct proportional control as illustrated in the schematic renders the system dynamics unstable and destroys the rotational spiral regardless of positive or negative feedback. Reproduced from [SS08] with permission.

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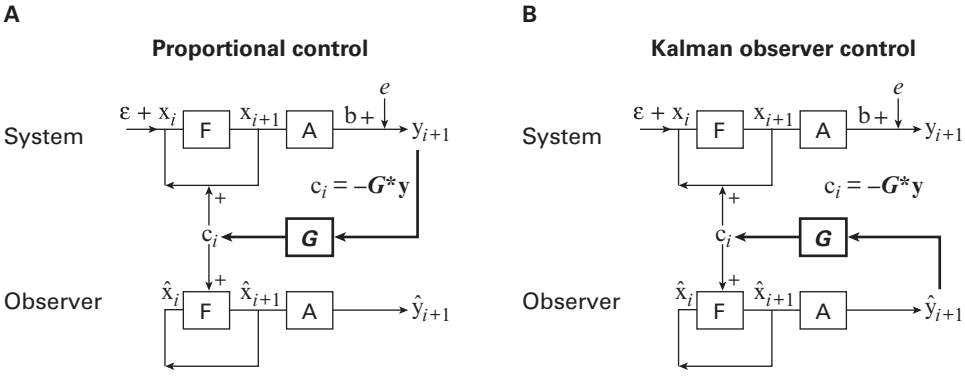


Figure 6.22
Schematic of the use of a Kalman filter observer in control. A illustrates the use of an observer to estimate state variables and parameters while control is derived from the actual noisy output y. The control vector c is applied to both the system and the observer. B shows the use of the observer system to calculate proportional control through an estimation of the observable $\hat{y}$. Again, the control vector is fed back into both the experimental system and the Kalman observer.

includes adding random noise e. We first assume that there is no model that observes this system, so we take measurement y and multiply it by G to generate control signal c, which is added to the dynamics F. This is the typical scenario where a noisy measurement is used in a *proportional control* feedback scheme. We examine the output of figure 6.21A and, for suitable initial conditions, we observe spiral waves, a snapshot of which is shown in figure 6.21B. For relatively low noise, and negative feedback ($G < 0$), we can observe slight slowing of the oscillations recorded from one node in figure 6.21C. If the strength of the negative feedback is increased, the oscillations are stopped completely, as shown in figure 6.21D. There is actually little safety margin between slowing the system and crushing the waves. In figure 6.21E, we use positive feedback ($G > 0$) to speed up the waves in low noise. It is always easier to speed waves up than to slow them down, whether in such models or in real experiments (e.g., [RSG05]). In high noise, the situation looks bleak. Whether one applies positive (figure 6.21F) or negative feedback (figure 6.21G), the system loses stability in the presence of feedback and the oscillations are destroyed.

Now let's introduce an *observer* system, as shown in figure 6.22. There are two general control schemes. One can keep the proportional control scheme as shown earlier, but observe the dynamics using an observer system. The control signal c is pumped into both the original system and the observer system. *Kalman filters really are filters*. They can extract from corrupted signals a signal reflective of the dynamics that the Kalman filter seeks to emulate. So a better way to reconstruct the control signal might be to calculate the control signal from the observer system, taking advantage of its filtering properties, and pump this signal into both the real and the observer system, as shown in figure 6.22B.

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We should be very clear about several things. If a simple system has low noise, then one has a better chance to effectively use the measured output in proportional control feedback scenarios to modulate the system (figure 6.21C and 6.21E). If all you want to do is to stop the activity of an excitable system, you can deliver a large current and stop the activity, as shown in figure 6.21D. In the brain, seizures can be stopped, and fibrillation in the heart, by delivering a large enough current. *You don't need a Kalman filter for mindless wave crushing.* But if you are in a regime where the amount of noise (or uncertainty in your model) is sufficiently large that adding feedback control destabilizes the system, you can improve the situation with model-based observation. In particular, the best scenario is when control is calculated from a Kalman observer when the observer system is built from a reasonable model. In figure 6.23 (plate 8), the left side shows a region where there is sufficient noise so that proportional control with positive gain calculated from the noisy measurement destabilizes the system (left column), whereas control calculated from the Kalman observer is capable of speeding up the system without destabilization.

An analogous situation is shown in figure 6.24 (plate 9). Here again, use of a proportional control signal with negative gain calculated from the noisy measurements destabilizes the system (left), whereas use of the Kalman observer calculated control leads to controlled slowing of the system (on the right).

In both of these numerical examples, we calculate the amount of control energy expended to achieve a given level of control. As one might imagine, when the control law $c = Gy$ is buffeted constantly by the noisy output y, this leads to wild excursions in the modulated activity, and the energy pumped into the system is large. When the control law is calculated from the Kalman observer, an amount of control energy more pertinent to the control law is expended, and the energy used is considerably reduced (we will show that it can be exponentially reduced in chapter 12). Although the control energy is not as critical when dealing with numerical simulations, it is crucial when dealing with brains [FSBG$^+$07, FGR$^+$07].

Critical in demonstrating the results in figures 6.23 and 6.24 was the use of *covariance inflation* to tune the measurement uncertainty used in the covariance matrix of the estimated state. Recall that Julier and Uhlmann [JU97a, JU97b] employed an arbitrary parameter on the spread of their sigma points (chapter 2). Here we follow a similar strategy as proposed by Hunt and colleagues [HKS07]. Without employing such adjustable covariance inflation, our results would have been unstable.

The philosophy here is that, in a perfect world, you would create a narrow uncertainly ball of sigma points that would enclose within the true state, and track this true state accurately as the model iterated. But the danger is that if you lose track of the system, you have no idea how to recover the trajectory. The alternative is to expand the uncertainty ball. The extreme of this would be that in a navigation scenario, were the uncertainty ball as large as the planet, you would never lose track of your position, but your uncertainly over where you are would encompass the entire planet! In the results here, we arbitrarily tuned a covariation parameter (to be discussed in this chapter's exercises). Improving the way we

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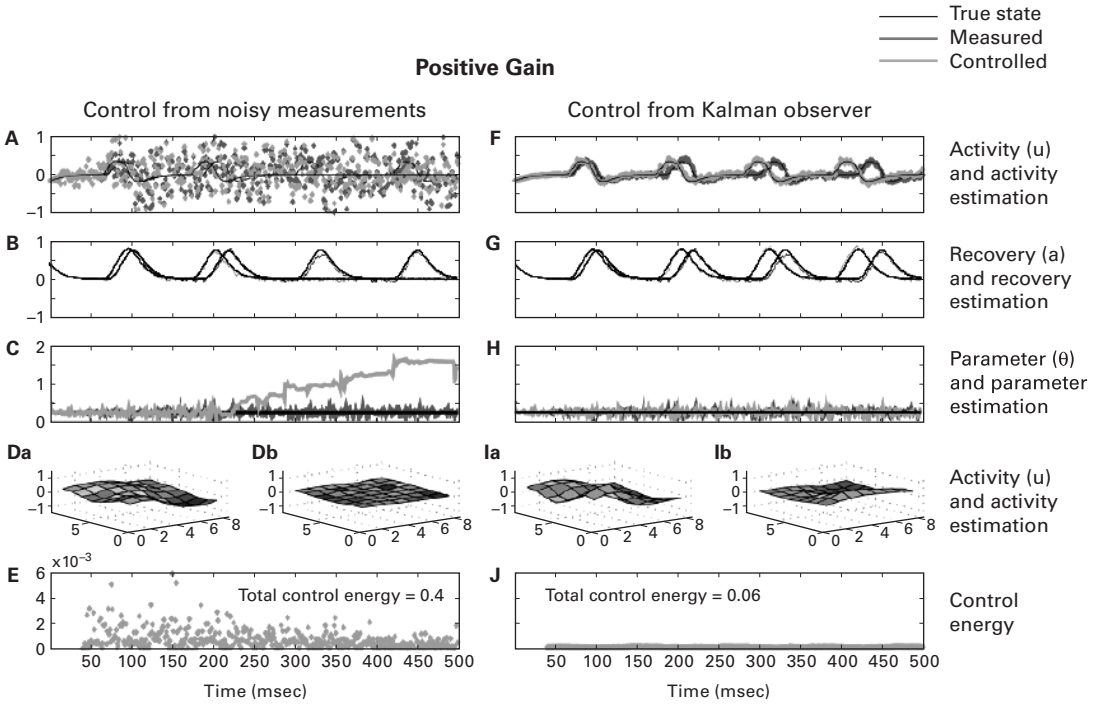


Figure 6.23 (plate 8)
Comparison of the use of positive gain proportional control calculated directly from high noise observations, left column A–E, and from the Kalman observer, right column F–J. (A) Output variable y_i from single element of grid, is the same as in figure 6.21F. The application of positive feedback gain at this high level of noise results in instability of the rotational oscillation, with loss of both the underlying oscillation u_i (black line) and measured observation y_i (red). (B) Without control being applied, the Kalman observer can accurately estimate (blue) the unobserved variable a (black). With this level of noise, when control is applied, not only is the observable a oscillation lost (red), but the true system state loses stability and the wave is destroyed after two oscillations (black). (C) Without control, despite the high noise level, the parameter θ (black) is accurately tracked (blue). When the stability of the dynamics is lost during control, the parameter θ is no longer accurately estimated (red). Da shows the variable y, and Db the estimated underlying activity $\hat{u}$. The snapshots in Da and Db are taken from the last frame in the 500 msec simulation, and the full simulation can be viewed as Supplementary Movie 1 from [SS08]. In E is shown the energy (sum of squared voltage) of the control vector c_i, and the total sum of all control vector energy for the 500 msec is 0.4 (arbitrary scale). F shows the estimated observable $\hat{y}$ without (blue) and with (red) control derived from the Kalman observer. (G) With Kalman observer control, the unobserved recovery variable a is now well tracked and the rotational spiral is stably increased in frequency, and in H, the parameter θ is now well tracked. Ia and Ib show the experimental variable y and estimated underlying variable $\hat{u}$. In J is shown that the control energy required is markedly reduced, with total control energy 0.06 (arbitrary scale).

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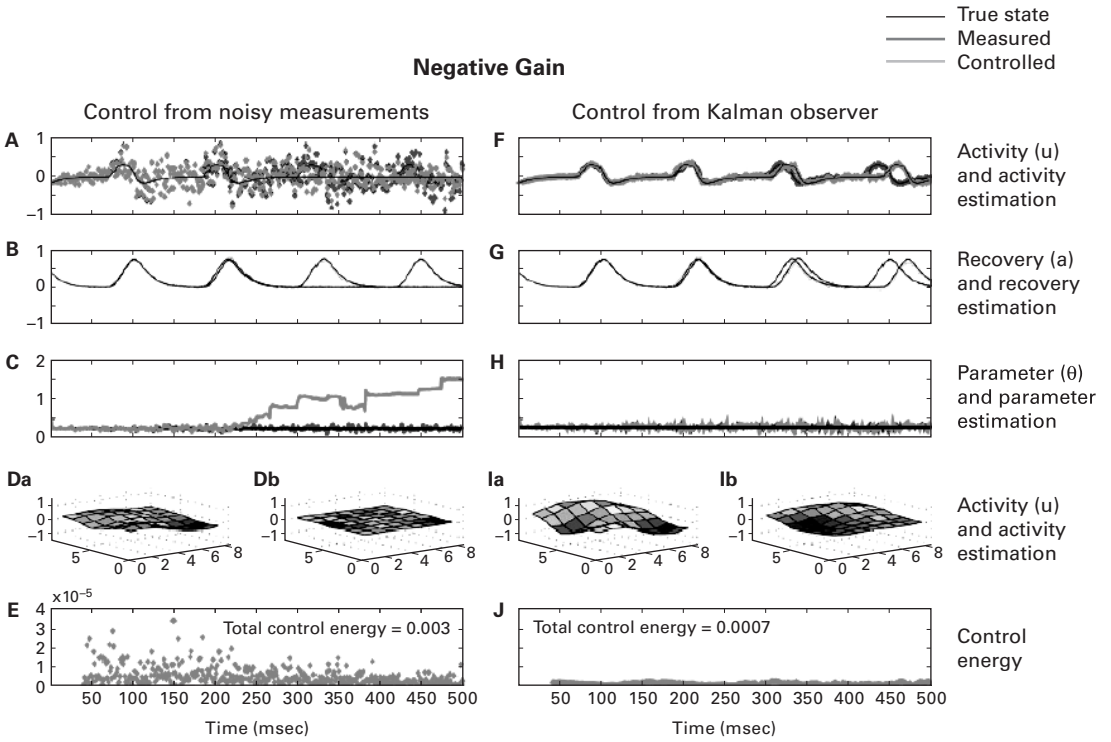


Figure 6.24 (plate 9)
Comparison of the use of negative gain proportional control calculated directly from high noise observations. In left column A–E, and from the Kalman observer, in right column F–J. (A) Output variable y_i from single element of grid, same as in figure 6.21G. The variables in this figure are similar to the description for figure 6.23. Note that with control derived from the Kalman observer (right column), negative gain in high noise conditions stably slows the frequency of the spiral, and requires considerably less control energy than the unstable case where direct proportional control is calculated from the noisy observations (left column).

handle covariation inflation adaptively is at the cutting edge of such algorithm design. We will return to this issue in chapter 8 on model inadequacy.

Note, also, that our Wilson-Cowan model had homogenous parameters—a rather ridiculous assumption if the underlying system did not (it did in this case). We will address this issue more realistically in chapter 8.

Our use of a fundamental model, the Wilson-Cowan equations, trusted that we knew a lot about the underlying system. The alternative is to use an empirical model based only on the data itself. This is a classic tussle between arrogance and ignorance. I am not sure what is the best way to handle such system tracking at present, and we will cover empirical modeling of these same dynamics in chapter 7. If you were designing a control system, hedging your bets is probably the safest route. Since each model, fundamental or empirical,

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can have prediction errors compared at each time step, one can run both in parallel if you have the computational overhead. If you don't this year, you might next year. And such parallel algorithms are natural for parallel computational implementation. Nevertheless, as we continue to learn more about the nervous system, our fundamental models will eventually overtake our empirical ones in control accuracy.

The preceding calculations implicitly employed direct injection of current to the elements of the discretized Wilson-Cowan equations. Subtleties of true electric field polarization of extended neurons, with the counter-fields generated within each neuron, and the ephaptic electrical crosstalk present between real neurons ([PBG+05]) will complicate the picture during experiments with real neuronal networks. Such complications may dictate that a more detailed model, incorporating the polarization characteristics of neurons as well as the electrically resistive properties of brain tissue within which the neurons are embedded, will be required to provide more accurate performance.

Exercises

6.1. Let's take the Wilson-Cowan equations as shown in (6.9) and work out some ways of calculating them.

• First let's set the parameters as

k = 0.05; B = 1000; K = 10; C = 15; tau = 0.01; Threshold = 0.1

with an integration time of delt = 0.001.
Use a 12×12 grid, and perhaps 200 iterations.

• Now let's do the full brute force version of these equations. You can set the initial conditions for u and a as

u0 = rand(n, p);

a0 = zeros(n, p);

where $n = p = 12$.

• Here is a code fragment that will get you started:

```
for step=1:numsteps
  for i=1:n
    for j=1:p
      integ=0;
      for a=1:n
        for b=1:p
          if (a==1|a==n)|(b==1|b==p);
           u0(a,b)=0; end %Neumann boundary conditions
```

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```
          if u0(a,b)>Threshold
            r=(i-a)^2+(j-b)^2;
            if Gaussian == FALSE%exponential
              integ=integ+K*exp(-k*sqrt(r));
            else%is Gaussian
              integ=integ+K*exp(-k*r);
            end
          end
        end
      end
   a1(i,j)=a0(i,j)+delt*(B*u0(i,j)-a0(i,j))/tau;
   u1(i,j)=u0(i,j)-delt*C*u0(i,j)-delt*a1(i,j)+delt*integ;
  end
  u0=u1;a0=a1;
  [YOUR FAVORITE PLOT ROUTINE HERE]
end
```

- Try replacing the initial conditions with a small bump, such as

```
u0=zeros(n,p);% this gives initial condition bump
  for i = 2
    for j = 3
      u0(i,j)=1;
    end
  end
```

- The above algorithm was accurate to the equations but a bit miserable in terms of efficiency. Here, courtesy of my colleague and collaborator Tim Sauer, is a much more compact and efficient version:

```
q=2; %or anything up to 12
for step=1:numsteps
  ue=[zeros(p,q) u0 zeros(p,q)]; %pad the grid with zeros
  ue=[zeros(q,n+2*q);ue;zeros(q,n+2*q)];
  integ=zeros(n,p);
  for i=-q:q  %check firing (i,j) steps away on grid
    for j=-q:q
      integ=integ+K*exp(-k*(i*i+j*j))* ...
      (ue(i+1+q:i+n+q,j+1+q:j+p+q)>Threshold);
    end
  end
  integ=integ-K*(u0>Threshold); %subtract self-input i = j = 0
  a1=a0+delt*(B*u0-a0)/tau;
  u1=u0-delt*C*u0-delt*a0+delt*integ;
  u0=u1;a0=a1;
  [YOUR FAVORITE PLOT ROUTINE HERE]
end%numsteps
```

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• Did you see any spirals? Probably not. Try these initial conditions and set the grid size to n = 8; p = 8;

```
u0=[-0.1637  -0.2244  -0.1982  -0.1410  -0.1029  -0.0811  -0.0656  -0.0396
    -0.2593  -0.3580  -0.3021  -0.2008  -0.1321  -0.1028  -0.0840  -0.0516
    -0.2444  -0.3386  -0.2746  -0.1726  -0.0950  -0.0742  -0.0637  -0.0417
    -0.0055  -0.0383   0.0012  -0.0657  -0.0395  -0.0388  -0.0387  -0.0270
     0.2361   0.3685   0.2616   0.2237   0.0279  -0.0174  -0.0239  -0.0183
     0.4698   0.7002   0.5627   0.2726   0.0939  -0.0071  -0.0171  -0.0135
     0.3618   0.5613   0.4323   0.2754   0.0673  -0.0056  -0.0130  -0.0104
     0.2442   0.2669   0.2571   0.1067   0.0152  -0.0049  -0.0077  -0.0061
   ];

a0=[ 0.4104   0.5339   0.4345   0.2786   0.1904   0.1454   0.1157   0.0694
     0.7058   0.9183   0.6974   0.4049   0.2427   0.1821   0.1465   0.0893
     0.9263   1.1846   0.8515   0.4104   0.1757   0.1283   0.1085   0.0707
     1.0750   1.4204   1.0146   0.4695   0.0921   0.0646   0.0636   0.0444
     1.0367   1.4612   0.9507   0.4993   0.0532   0.0309   0.0380   0.0293
     0.7878   1.1079   0.7914   0.3707   0.0468   0.0205   0.0267   0.0213
     0.4156   0.5573   0.4218   0.1886   0.0255   0.0150   0.0202   0.0162
     0.1386   0.1823   0.1406   0.0437   0.0083   0.0090   0.0119   0.0095
   ];
```

Setting different initial conditions, or sequences of initial conditions, produces a wide range of dynamics from these equations.

6.2. Now let's explore how to modify the Voss algorithm, presented in the exercises in chapter 5, so that we can apply it spatiotemporally.

• We add covariance inflation to the ut subroutine.

```
ip=.0001;% ep is the inflation parameter
Pxx=(Pxx+Pxx')/2+ip*eye(size(Pxx));
```

The rest of the ut subroutine is remarkably unchanged.

• The observation function is easy:

```
function r=kalmanwc_obs(x,dq)
global K C B tau f g dt k nn
r=x(dq+1:f*g+dq,:);%just the u's
```

• The integrator function is a bit more complex.

```
function r=kalmanwc_int(x,z)
global K C B tau f g dt k nn

u = reshape(x(1:f*g),f,g);
a = reshape(x(f*g+1:2*f*g),f,g);
```

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```
q=2;
p=[K,C,B,tau,z];
ue=[zeros(g,q) u zeros(g,q)];
ue=[zeros(q,f+2*q);ue;zeros(q,f+2*q)];
integ=zeros(f,g);
for i=-q:q                      %check firing (i,j) steps away on grid
 for j=-q:q                     %add to input
  integ=integ+p(1)*exp(-k*(i*i+j*j))*(ue(i+1+q:i+f+q,j+1+q:j+g+q)>p(5));
 end
end
integ=integ-p(1)*(u>p(5));    %subtract self-input i = j = 0
udot=(-p(2)*u-a+integ);       %ODE right hand side
adot=(p(3)*u-a)/p(4);         %for Wilson-Cowan
r=[udot(:);adot(:)];%a column vector with both variables
```

- The vectorized integrator called from the ut subroutine is now

```
function r=kalmanwc_fct(dq,x) %x here is [z ; u ; a]
%original z will be returned!
global K C B tau f g dt k nn

dT=nn*dt;

p=x(1:dq,:); %strips out p from first row
xn=x(dq+1:size(x(:,1)),:);
for n=1:nn  %Runge-Kutta order 4
  k1=dt*fc(xn,p);
  k2=dt*fc(xn+k1/2,p);
  k3=dt*fc(xn+k2/2,p);
  k4=dt*fc(xn+k3,p);
  xn=xn+k1/6+k2/3+k3/3+k4/6;
end
r=[p; xn]; %returns original p

function w=fc(x,p1);
global K C B tau f g dt k
[Rows,Columns]=size(x);
w=zeros(Rows,Columns);
for NumberOfColumns = 1:Columns
  u = reshape(x(1:f*g,NumberOfColumns),f,g);
  a = reshape(x(f*g+1:2*f*g,NumberOfColumns),f,g);
  q=2;p=p1(:,NumberOfColumns);
  p=[1.38;3;10;4.85;p];
  ue=[zeros(g,q) u zeros(g,q)];
  ue=[zeros(q,f+2*q);ue;zeros(q,f+2*q)];
  integ=zeros(f,g);
  for i=-q:q                    %check firing (i,j) steps away on grid
    for j=-q:q                  %add to input
```

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```
      integ=integ+p(1)*exp(-k*(i*i+j*j))*(ue(i+1+q:i+f+q,j+1+q:j+g+q)>p(5));
    end
  end
  integ=integ-p(1)*(u>p(5));   %subtract self-input i = j = 0
  udot=(-p(2)*u-a+integ);      %ODE right hand side
  adot=(p(3)*u-a)/p(4);        %for Wilson-Cowan
  w(:,NumberOfColumns)=[udot(:);adot(:)];%column vector with both variables
end
```

• Finally, here is a simplified version of the main script to control the Wilson-Cowan equations using an observer system. The full version of the program is archived with [SS08]. For initial conditions, use the matrices at the end of exercise 6.1.

```
%Kalman filter applied to Wilson-Cowan dynamics
%Steven Schiff and Tim Sauer, 2007-2009
%Simplified from Original Archived with Journal of Neural Engineering Paper
%"Kalman Filter Control of Spatiotemporal Cortical Dynamics"
% We have followed structure from Voss et al 2004
% Octave and Matlab compatible
clear all;close all
TRUE = 1; FALSE = 0;
global K C B tau f g dt k nn
%***SELECT the Gain, NoiseFactor, to reproduce calculations in paper
  NoiseFactor = 10
  EnergyPlotYAxis=0.00004
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%n.b.: increasing negative gain will quench the waves
%For instance, try Gain=-0.05 to see this clearly.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% p=[K,C,B,tau,Threshold]
  K=1.38;C=3;B=10;tau=4.85;k=0.91;%5 parameters plus 2 variables each cell
%Dimension of Grid
  f=8;g=f;
% Dimensions: (dq for param. vector, dx augmented state, dy observation)
  dq=1; dx=dq+2*f*g; dy=f*g;
fct='kalmanwc_fct'; % this is the model function F(x) used in filtering
obsfct='kalmanwc_obs';  % this is the observation function G(x)
N=500% number of data samples
dt=0.06;nn=1;dT=nn*dt;  %Step size dt
MaxPlotEnergy = eps;
ControlTime = 40
EnergyThreshold = 4
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
for WithControl = 1:2
  if WithControl == 1
    Gain = 0%get baseline %use with noise 10 - shows advantage of Kalman Observer
```

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```
  else
    Gain = -0.0006 %can set this + or -
  end%otherwise use Gain as set
% Initial conditions for estimation
  MSEsum = 0; %initialize
  uEnergy_y_sum = 0; uEnergy_yhat_sum = 0;
  u = load('-ascii', 'u_initial_8x8_spiral.txt');
  a = load('-ascii', 'a_initial_8x8_spiral.txt');
  clear x0
  x0(:,1)=[u(:); a(:)]; %1 big column vector of all variables for initial time
% External input, estimated as parameter p(5) later on:
  z=ones(1,N)*0.24; %z=[1:N]/50*2*pi; %z=[0.24+0.00*sin(z)];
  x(:,1)=[z(1,1); x0]; % augmented state vector (notation different from paper)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
t=1; %initialize
  xhat(:,1)=x(:,1); % first guess of x_1 set to observation
  xhat(1,1)=0.55;  % set first guess of first parameter arbitrarily
  yhat= xhat(dq+1:dq+f*g,1);
  Q=.0001  % process noise covariance matrix
  errors=zeros(dx,N);
  Energy=zeros(1,N);
  tempx=x(:,1)*ones(1,N);
  R=.2^2*cov(kalmanwc_obs(tempx(:,1:N),dq)'); % observation noise cov matrix
  R=R+eps*eye(size(R));
  Pxx(:,:)=blkdiag(Q,R,R); % need one copy of Q for each tracked parameter
  randn('state',3);
  y=feval(obsfct,x(:,1),dq)+sqrtm(R)*randn(dy,t); % noisy data
  randn('state',0); %Not ysing this - set state above
  Energy_y(t)   = sum( y(:, t) .* y(:, t) ); % y
  Energy_yhat(t) = sum( yhat(:, t) .* yhat(:, t) ); % y
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
  for t = 1:N-1
    if t>1
      xx=x0(:,t-1); %Pick column (at start, only 1 column of initial conditions)
      if t>=ControlTime
        uVector_yhat = Gain * yhat(:,t-1);
        uEnergy_yhat(t) = uVector_yhat' * uVector_yhat;
        ControlVector_yhat = [uVector_yhat ; zeros(f*g,1)];
        xx = xx + ControlVector_yhat;
        xhat(:,t-1) = xhat(:,t-1) + [0 ; ControlVector_yhat];
      else
        uEnergy_yhat(t) = 0; %no control yet
      end
      for i=1:nn %nn determines how many times to loop the RK - can use 1
        k1=dt*kalmanwc_int(xx,z(:,t-1));%feed a col of x and a value of z
        k2=dt*kalmanwc_int(xx+k1/2,z(:,t-1));
```

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```
        k3=dt*kalmanwc_int(xx+k2/2,z(:,t-1));
        k4=dt*kalmanwc_int(xx+k3,z(:,t-1));
        xx=xx+k1/6+k2/3+k3/3+k4/6;
      end;
      x0(:,t)=xx;
      x=[z(1:t); x0(:,1:t)]; % augmented state vector
      R=.2^2*cov(kalmanwc_obs(x(:,1:t),dq)'); % observation noise cov matrix
      R=R+eps*eye(size(R));
      y=feval(obsfct,x,dq)+NoiseFactor*sqrtm(R)*randn(dy,t); % noisy data
      [xhat(:,t),Pxx]= ...
                kalmanwc_ut(xhat(:,t-1),Pxx,y(:,t),fct,obsfct,dq,dx,dy,R);
      Pxx(1,1)=max(Pxx(1,1),Q); % keep the parameter cov from fading to 0
      yhat(:,t)= xhat(dq+1:dq+f*g,t);
      Energy_y(t) = sum( y(:, t) .* y(:, t) ); %the rms at time t
      Energy_yhat(t) = sum( yhat(:, t) .* yhat(:, t) ); % y
      errors(:,t)=sqrt(diag(Pxx(:,:)));
    end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
     subplot(5,4,1:4)
      hold on;
      plot(x(dq+1,1:t)','k','LineWidth',.5);
      if Gain == 0
        plot(t,y(1,t)','bd','MarkerEdgeColor','blue', ...
          'MarkerFaceColor','blue','MarkerSize',5);
      else
        plot(t,yhat(1,t)','rd','MarkerEdgeColor','red', ...
          'MarkerFaceColor','red','MarkerSize',5);
      end
      xlabel('t','fontsize',14);
      axis([1 N -1 1])
      title('Control from est observations yhat','fontsize',14)
      ylabel('x state, y obs','fontsize',14);
      drawnow
     subplot(5,4,5:8)
      plot(x(dy+2,1:t)','k','LineWidth',.5);%unobserved variable
      hold on
      plot(xhat(dy+2,1:t)','r','LineWidth',.1);%est unobserved variable
      axis tight
      axis([1 N -1 1])
      ylabel('a, est a','fontsize',14)
     subplot(5,4,9:12)
      hold on
      plot(xhat(1,1:t),'r','LineWidth',2);
      plot(x(1,1:t),'k','LineWidth',2); %parameter
      axis([1 N 0 2.0])
      ylabel(['thr',', ',' est ','thr'],'fontsize',14);
```

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```
    subplot(5,4,13:14)
     u = reshape(y(:,t),f,g);%Display noisy observations
     surf(u);title('u','fontsize',12);axis([0 f 0 g -1.5 1.5])
     drawnow
    subplot(5,4,15:16)
     uhat = reshape(xhat(dq+1:f*g+dq,t),f,g);
     surf(uhat);title('uhat','fontsize',12);axis([0 f 0 g -1.5 1.5])
     drawnow
    subplot(5,4,17:20)
     if t>ControlTime
       if uEnergy_yhat(t)>MaxPlotEnergy; ...
           MaxPlotEnergy = uEnergy_yhat(t); end %store max graph value
       uEnergy_yhat_sum = uEnergy_yhat_sum + uEnergy_yhat(t);
       if Gain == 0
         plot(t,uEnergy_yhat(t),'bd','MarkerEdgeColor','blue', ...
           'MarkerFaceColor','blue','MarkerSize',5);
       else
         plot(t,uEnergy_yhat(t),'rd','MarkerEdgeColor','red', ...
           'MarkerFaceColor','red','MarkerSize',5);
       end
       hold on
       xlabel(['total control engy =...
                                ',num2str(uEnergy_yhat_sum)],'fontsize',14)
       ylabel('u Energy','fontsize',14);
       axis([1 N 0 MaxPlotEnergy])
     end
  end%t
end%WithControl
```

6.3. So can you substitute Fitzhugh-Nagumo dynamics for Wilson-Cowan in the preceding example? It is really a replacement of the dynamics at each node in the network—all the rest stays the same. The parameters are different.

- *Hint:* Try setting global variables to

global A B C f g dt K k nn

and use parameter settings as

$K = 1.38; A = 0.7; B = 0.8; C = 3; k = 0.91;$

Later, we will explore using the one model to track a different (wrong) model. It is not that unreasonable a suggestion. In real life, all of these models are *wrong* in that they do not represent nature correctly, but that is a subject for a later chapter.

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